



EXAMINATION NO.: _____
THE MALAWI NATIONAL EXAMINATIONS BOARD

MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

SAMPLE PAPER
AGRICULTURE

Subject Number: M012/I

Time Allowed: 2 hours

PAPER I
(100 marks)

Instructions

1. This paper contains 12 printed pages. Please check.
2. This paper has two sections. **A** and **B**.
Section **A** has ten questions and
Section **B**, three questions.
3. Answer **all** questions in the spaces provided
4. Write your Examination Number on **all** pages.
5. In the table provided on this page **tick** against the question number you have answered
6. At the end of the examination, hand in your paper to the invigilator

Question Number	Tick if answered	Do not write in these columns		Marker's Name
1				
2				
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12				
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SECTION A: 70 Marks

(Answer all the questions in the spaces provided)

- I. a. Give any two forms of soil degradation.

(2 marks)

- b. Explain any two ways in which poor farming practices encourage water loss.

(4 marks)

- c. State any two ways in which deforestation causes climate change

(2 marks)

- d. Explain any two ways of modifying acidic soils to suit maize Production.

(4 marks)

Continued/...

2. a. List any three ways of randomizing the treatments in an experiments

(3 marks)

- b. Explain any two ways in which agricultural research help to support the growing population in Malawi.

(4 marks)

3. **Table 1** shows results of an experiment on effect of fertilizer rate on maize yield. Use it to answer the questions that follow:

Table 1

Fertilizer rate (kg/ha)	Yield (kg)
0	1000
100	1700
200	3000
300	5000
400	7000
500	7500
600	7600

- (a) Plot a graph of yield against fertilizer rate on **page 4**. (4 marks)

Continued/...

3. (Continued)

[illegible]

Continued/...

3. (Continued)

- b. From the graph what is the best fertilizer rate that can be recommended for growing maize?

(1 mark)

- c. Give a reason for the answer in 5a.

(2 marks)

- d. State the economic principle which has been illustrated on the graph.

(1 mark)

- e. What is the importance of analyzing data in this experiment?

(1 mark)

4. Explain any **two** conditions necessary for setting up the following irrigation systems:

- a. Furrow irrigation

(2 marks)

- b. Sprinkler irrigation

(2 marks)

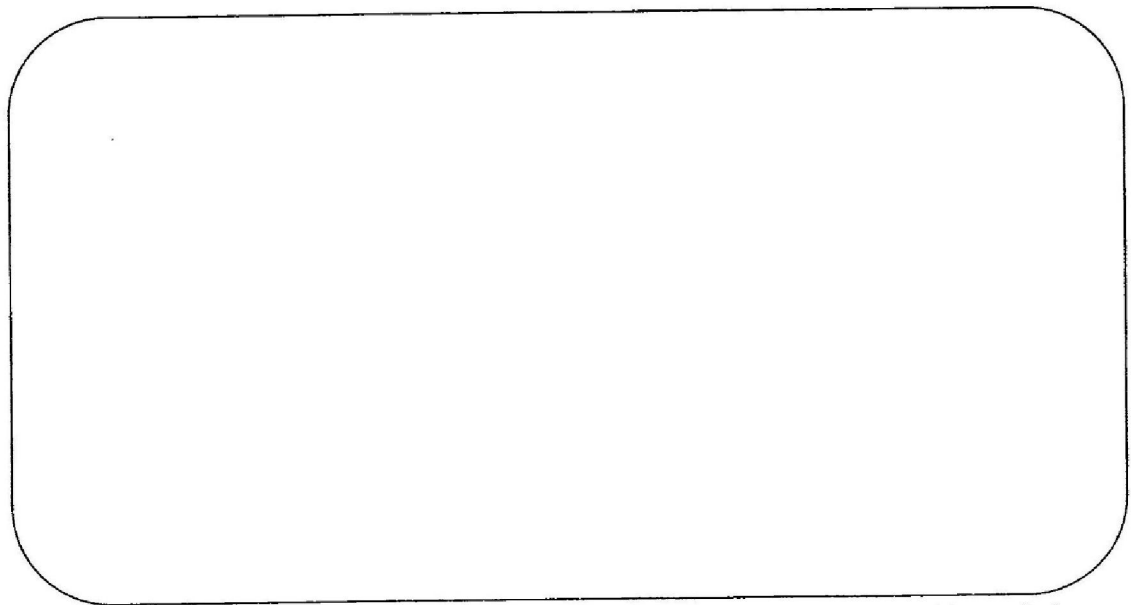
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5. **Table 1** shows different prices of tomato and their corresponding levels of demand. Use it to answer the question that follows:

Price (MK/tomato)	Quantity demanded
5	120
10	100
15	80
20	60
25	40
30	20

Suppose the price for each tomato changed from K10 to K15

- a. calculate price elasticity of demand



(4 marks)

- b. List any three major factors of production.

(3 marks)

Continued/...

6. State any **two** advantages and **two** disadvantages of using vegetative planting materials.

Advantages

(2 marks)

Disadvantages

(2 marks)

7. A maize plant is showing chlorosis starting from the tip of the leaf and stunted growth. Use the information to answer the questions that follow:

- a. Identify the nutrient that the maize plant is lacking.

(1 mark)

- b. State any **two** ways of correcting the deficiency symptoms.

(2 marks)

Continued/...

7. (Continued)

- c. Give any **two** ways in which the nutrient is being depleted from the soil.

(2 marks)

8. a. List any **two** major plant nutrients.

(2 marks)

- b. Describe any **three** ways of maintaining soil fertility.

(4 marks)

9. **Figure 1** is a diagram of livestock parasite. Use it to answer the questions that follow:



Figure 1

- a. Identify the parasite.

(1 mark)

Continued/...

9. (Continued)

- b. Name any **two** farm animals that are affected by the parasite.

(2 marks)

- c. State any **two** ways of controlling the parasite.

(2 marks)

10. (a) Name any **two** types of broiler feed.

(2 marks)

- (b) List any **three** diseases of chickens.

(3 marks)

- (c) Explain any **one** way of controlling wild cat.

(4 marks)

Continued/...

SECTION B: (30 marks)

Answer all the three questions in this Section

11. Explain any five advantages of mixed cropping.

(10 marks)

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END OF QUESTION PAPER

NB: This paper contains 12 printed pages.